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SYMBIOSIS

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Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy

constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement:

Develop a Java application demonstrating the use of default, parameterized, and copy constructors

for object initialization and data management.

Learning Objectives:

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory:

Constructors are special member functions that are automatically invoked when an object is created.

Java supports default constructors provided by the compiler, parameterized constructors for

initializing objects with user-defined values, and copy constructors (implemented manually) for

creating duplicate objects. Constructors improve code readability and simplify object initialization

while supporting object-oriented programming principles.

Exercise:

1. Create a Student class using default and parameterized constructors to initialize student name and roll Number.

Program:

```
File Edit Selection View Go Run Terminal Help projects
J Student.java 1 ●
helloworld > J Student.java > ...
1  import java.util.*;
2  public class Student {
3      String name;int Roll;
4  Student(String n,int roll)    //parameterized constructor
5  {
6      name=n;
7      Roll=roll;
8  }
9  Student()    //default constructor
10 {
11     name="Ayush";
12     Roll=58;
13 }
14 void display()
15 {
16     System.out.println("Name:"+name);
17     System.out.println("Roll number:"+Roll+"\n");
18 }
Run main | Debug main | Run | Debug
19 public static void main(String args[])
20 {
21     String name;int roll;
22     Scanner in=new Scanner(System.in);
23     System.out.println(x: "Enter name:");
24     name=in.next();
25     System.out.println(x: "Enter Roll number:");
26     roll=in.nextInt();
27     Student a=new Student(name,roll);
28     Student b=new Student();
29     a.display();
30     b.display();
31 }
32 }
```

Output:

```
PS C:\Users\Ayush\projects> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -cp 'C:\Users\Ayush\workspaceStorage\44eea058444\Student\Student.java' Student
Enter name:
Virat
Enter Roll number:
18
Name:Virat
Roll number:18

Name:Ayush
Roll number:58
```

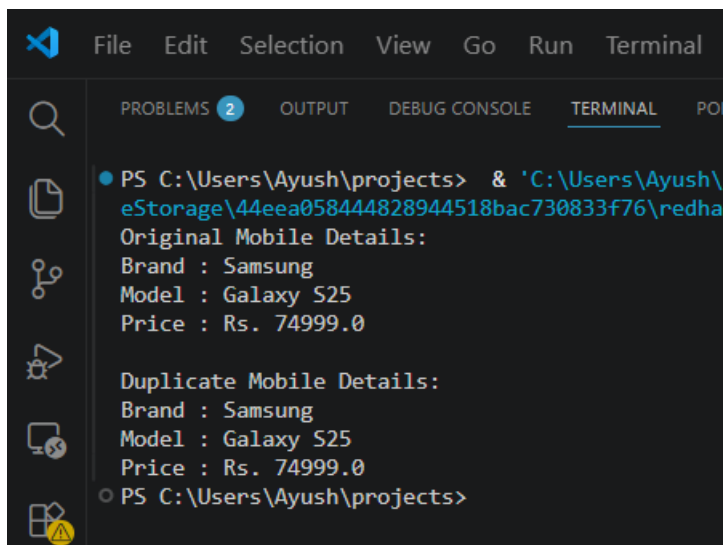
2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile

details and create duplicate object records.

PROGRAM:

```
File Edit Selection View Go Run Terminal Help
J MobileInventory.java
helloworld > J MobileInventory.java > Java > Mobile
1 class Mobile {
2     String brand;
3     String model;
4     double price;
5
6     // Default Constructor
7     Mobile() {
8     }
9
10    // Copy Constructor
11    Mobile(Mobile m) {
12        brand = m.brand;
13        model = m.model;
14        price = m.price;
15    }
16
17    void display() {
18        System.out.println("Brand : " + brand);
19        System.out.println("Model : " + model);
20        System.out.println("Price : Rs. " + price);
21    }
22 }
23
24 public class MobileInventory {
25     Run main | Debug main | Run | Debug
26     public static void main(String[] args) {
27
28         Mobile m1 = new Mobile();
29         m1.brand = "Samsung";
30         m1.model = "Galaxy S25";
31         m1.price = 74999;
32
33
34         Mobile m2 = new Mobile(m1);
35
36         System.out.println(x: "Original Mobile Details:");
37         m1.display();
38
39         System.out.println(x: "\nDuplicate Mobile Details:");
40         m2.display();
41     }
42 }
43
```

Output:



The screenshot shows a Visual Studio Code interface with a terminal window open. The terminal displays the following command and output:

```
PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\
eStorage\44eea058444828944518bac730833f76\redha
Original Mobile Details:
Brand : Samsung
Model : Galaxy S25
Price : Rs. 74999.0

Duplicate Mobile Details:
Brand : Samsung
Model : Galaxy S25
Price : Rs. 74999.0
PS C:\Users\Ayush\projects>
```

The terminal window has a dark theme and includes a sidebar on the left with icons for search, file explorer, and other VS Code features. The top menu bar shows 'File', 'Edit', 'Selection', 'View', 'Go', 'Run', and 'Terminal'.